REAL ESTATE BUYER SEGMENTATION & INVESTMENT PROFILING

Machine Learning Project Report | Corrected 150-Record Version

Project purpose

This project demonstrates how buyer information can be explored and grouped to understand different real-estate customer profiles and investment preferences.

Dataset integrity statement

The dataset used for this version contains 150 synthetic buyer records. These records were created for learning and demonstration. They are not actual customers, transactions, or proprietary Parcl data.

Prepared for internship project submission

1. Executive Summary

This project presents a small end-to-end demonstration of real-estate buyer profiling. It uses a synthetic dataset of 150 buyer records and explores buyer type, region, purchase purpose, financing status, referral channel, income bracket, property budget, units of interest and satisfaction. The goal is to identify patterns that can help explain differences between buyer groups. The results are exploratory and should not be treated as estimates of the real property market.

2. Business Problem

Real-estate businesses work with buyers who have different needs. Some buyers seek a primary residence, while others may be interested in investment properties or second homes. Buyers may also differ by budget, region, financing needs and buyer category. Understanding these differences can help businesses plan more relevant communication and customer support.

3. Project Objectives

- Explore the characteristics of a sample of real-estate buyers.
- Examine buyer categories, geographic regions and purchase purposes.
- Review budget, financing, units of interest and satisfaction patterns.
- Apply a machine-learning clustering approach to explore similar buyer profiles.
- Present key information through an interactive Streamlit dashboard.
- Explain the limitations of synthetic demonstration data.

4. Dataset Description

The dataset contains 150 rows and 12 columns. The fields are client_id, date_of_birth, gender, client_type, region, acquisition_purpose, loan_applied, referral_channel, income_bracket, property_budget_usd, units_interested and satisfaction_score. The buyer categories include Individual, Corporate, Institutional, High Net Worth and International. Purchase purposes include Investment, Primary Residence and Second Home.

5. Data Quality Checks

A programmatic check of the 150-record CSV found 150 unique client IDs, no duplicate client IDs and no missing values in the 12 columns. Date of birth is available as a source field; age can be derived from it when required. These checks describe the supplied CSV file and do not certify every possible issue in the deployed application.

6. Exploratory Data Summary

The 150 records include 67 buyers whose stated purpose is Investment, 63 whose purpose is Primary Residence and 20 whose purpose is Second Home. Buyer categories include 60 Individual, 35 Corporate, 24 Institutional, 20 High Net Worth and 11 International records. The regional distribution is 36 North America, 35 Latin America, 33 Europe, 23 Asia Pacific and 23 Middle East records. Loan status is 71 Yes and 79 No.

7. Numerical Summary

The mean property budget in the supplied synthetic dataset is approximately USD 1,713,133, and the median is USD 1,281,125. The average number of units of interest is approximately 3.13. The average satisfaction score is approximately 3.70 out of 5. These are descriptive statistics for this synthetic sample only; they are not market benchmarks.

8. Machine-Learning Approach

The project uses buyer segmentation as an unsupervised-learning problem. A clustering method groups records with similar characteristics. Categorical fields such as buyer type, region, purchase purpose and referral channel

need suitable encoding before they can be used in a numerical model. Numerical fields such as budget, units of interest and satisfaction can be scaled so that variables with larger numeric ranges do not dominate simply because of their units. Cluster labels are descriptive interpretations of patterns; they are not verified identities or ground-truth buyer classes. The current dashboard should be treated as the presentation of the running application output.

9. Dashboard

The Streamlit dashboard is the interactive interface for exploring the project. It displays the buyer count and visual summaries from the dataset. Users can use the displayed charts and information to review buyer characteristics and patterns. The deployed dashboard should be opened during the project explanation video so the evaluator can see the working application.

10. Business Use and Recommendations

• Consider separate communication for buyers seeking a primary residence and buyers interested in investment.
• Review regional patterns before planning location-specific campaigns.
• Use budget and financing information to understand possible support needs.
• Test marketing ideas and measure qualified leads, conversions and customer satisfaction before making business decisions.

11. Limitations and Responsible Use

All records in this dataset are synthetic. They cannot establish actual buyer behaviour, market demand, property values or investment performance. The sample is small and designed for demonstration. Before any business use, the workflow should be tested with representative, lawfully obtained and appropriately consented data. Privacy, fairness, data quality, model stability and performance should also be reviewed.

12. Conclusion

This project demonstrates a practical workflow for exploring real-estate buyer information, applying buyer segmentation concepts and presenting the analysis in a Streamlit dashboard. It provides learning experience in data handling, exploratory analysis, machine-learning concepts and dashboard development. The findings are illustrative because the dataset is synthetic.

Dataset summary table

Measure	Value
Records	150
Columns	12
Unique client IDs	150
Duplicate client IDs	0
Missing values	0
Average property budget	USD 1,713,133
Median property budget	USD 1,281,125
Average units interested	3.13

Measure	Value
Average satisfaction	3.70 / 5

Important consistency note: This corrected report uses the 150-record dataset currently reflected by the deployed dashboard. It replaces the older report version that described a different 1,500-record dataset.